

LIQUIDMIND®.AI

Product Portfolio — Agentic AI for International Trade

Built on AWS with Anthropic Claude frontier models on Amazon Bedrock · NVIDIA Inception · DPIIT Recognised

LIQUIDMIND®.AI offers three flagship Agentic AI SaaS products purpose-built for Indian EXIM trade — directly relevant to steel and commodity trade compliance across the Ministry of Steel ecosystem, including SAIL, MSTC, and NMDC.



TradeGuard AI

Shipping Bill & Invoice Cross-Validation Engine

40+

fields cross-checked

< 5 sec

upload to full report

95%

accuracy with audit trail

Upload a Shipping Bill and Commercial Invoice; TradeGuard cross-checks 40+ critical fields — HSN codes, FOB values, exporter details, IEC numbers, port codes, unit of quantity — and flags every mismatch, warning, and match with colour-coded severity before submission to Customs or DGFT. For steel sector enterprises, it prevents the field-level discrepancies that trigger customs holds, drawback rejections, and RoDTEP claim losses.

- **40+ fields:** Exporter details, HSN codes, FOB value, port codes, IEC numbers, quantities, currency, shipping mode extracted automatically from both documents
- **Semantic port code matching:** INNSA1, NHAVA SHEVA, and JNPT resolved as the same port — zero false positives from format differences across ICEGATE, DGFT, and bank documents
- **Colour-coded severity:** Critical mismatches, warnings, and matches visually differentiated; every flagged field traceable to its source location in the original document
- **Human-in-the-loop review:** AI flags issues; the operator makes the final call before any document proceeds
- **Complete audit trail:** Every cross-check logged with timestamps and field-level evidence for CAG and statutory audit requirements
- **DGFT & Drawback protection:** Catches the discrepancies that cause RoDTEP and Drawback claim rejections before they occur



Patram AI

Trade Document Intelligence & Compliance Q&A

190+

country frameworks

1.5 sec

sourced answers

24x7

no consultant needed

Upload any trade document — export policies, FTA agreements, compliance guides, regulatory notifications — and ask questions in plain English. Patram returns a precise answer citing the exact page and section in 1.5 seconds, across 190+ country regulatory frameworks. It eliminates the 4–6 hour manual research cycle that precedes every cross-border shipment decision.

- **Plain-English Q&A on any document:** Export policies, FTA texts, DGFT notifications, compliance guides, tariff schedules
- **Every answer sourced:** Responses cite the exact page and section — full traceability, zero guesswork

- **190+ country coverage:** US, EU, UAE, ASEAN, and emerging markets; trade policies continuously updated
- **Multi-modal document reading:** Understands text, tables, charts, and nested layouts inside complex PDF trade documents
- **Verified answers:** Built-in fact verification layer prevents hallucinations at the root



TariffIQ (HSN+ AI)

HS Classification & Duty Incentive Calculator

21,000+

HSN codes, semantic search

< 3 sec

8-digit ITC-HS + incentives

₹ per shipment

RoDTEP vs Drawback delta

Describe your product in plain English; TariffIQ returns the exact 8-digit ITC-HS code with a confidence score and simultaneously compares RoDTEP vs Duty Drawback rates — showing the exact rupee difference per shipment. Live BCD, IGST, and Cess rates and export policy status (Free, Restricted, Prohibited) delivered instantly.

- **8-digit ITC-HS classification:** Semantic search across 21,000+ HSN codes with 95% accuracy and a confidence score on every result
- **RoDTEP vs Drawback side-by-side:** Both incentive rates shown with a clear recommendation and exact rupee difference per shipment
- **Live duty rate lookup:** Real-time BCD, IGST, and Cess rates; rate notifications and budget changes reflected immediately
- **Export policy status:** Required licences identified before shipment commitment
- **Batch classification:** Applicable to high-volume MSTC and PSU commodity operations
- **Explainable AI output:** Regulatory citation for each classification decision



Steel Sector Use Case

Coking Coal Import Intelligence — SAIL Corporate Office

Developed with SAIL's Coal Import Group (CIG), this use case applies the full product suite to one of India's largest import programmes — coking coal:

- **TradeGuard AI:** Cross-validates the 15–40 trade documents per vessel — commercial invoices, bills of lading, certificates of sampling & analysis, customs filings — catching field-level mismatches worth lakhs before they trigger holds or demurrage
- **Patram AI:** Gives sourcing, trading, and compliance teams instant sourced answers on contract terms, FTA eligibility, and regulatory obligations across supplier geographies (Australia, USA, Russia, Mozambique)
- **TariffIQ:** Correct 8-digit CTH classification for coking coal grades and blends, with live BCD/IGST/Cess and duty-incentive comparison — eliminating re-assessment risk on high-value cargoes

The same suite extends to plant-level workflows across SAIL Rourkela, Durgapur, and IISCO Burnpur, and to MSTC's e-commerce, auction, and trade facilitation mandate.

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